



Proving Trigonometric Identities

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Learning Objectives

- Apply fundamental trigonometric identities (reciprocal, quotient, and Pythagorean) to simplify expressions
- Prove trigonometric identities by transforming one side to match the other
- Recognize and use co-function, even-odd, and sum/difference identities in proofs

Prove each of the following trigonometric identities by showing all algebraic steps that transform the left-hand side into the right-hand side.

1. Prove the identity:

$$\sin(\theta) \cdot \cot(\theta) = \cos(\theta)$$

Answer: _____

2. Prove the identity:

$$\frac{1 - \cos^2(\theta)}{\sin(\theta)} = \sin(\theta)$$

Answer: _____

3. Prove the identity:

$$\sec(\theta) - \sin(\theta)\tan(\theta) = \cos(\theta)$$

Answer: _____

4. Prove the identity:

$$\frac{1 + \tan^2(\theta)}{\sec^2(\theta)} = 1$$

Answer: _____

5. Prove the identity:

$$\frac{\cos(\theta)}{1 - \sin(\theta)} = \frac{1 + \sin(\theta)}{\cos(\theta)}$$

Answer: _____

6. Prove the identity:

$$\tan(\theta) + \cot(\theta) = \sec(\theta)\csc(\theta)$$

Answer: _____



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7. Prove the identity:

$$\frac{\sin(\theta)}{1 + \cos(\theta)} + \frac{1 + \cos(\theta)}{\sin(\theta)} = 2\csc(\theta)$$

Answer: _____

8. Prove the identity:

$$(1 - \cos(\theta))(1 + \cos(\theta)) = \sin^2(\theta)$$

Answer: _____

9. Prove the identity:

$$\frac{\sec^2(\theta) - 1}{\sec^2(\theta)} = \sin^2(\theta)$$

Answer: _____

10. Prove the identity:

$$\csc(\theta) - \sin(\theta) = \cos(\theta)\cot(\theta)$$

Answer: _____





Encourage students to work on the more complicated side first and to convert all functions into sines and cosines when stuck.

Solutions

1. Prove the identity:

$$\sin(\theta) \cdot \cot(\theta) = \cos(\theta)$$

- Rewrite $\cot(\theta)$ as $\cos(\theta)/\sin(\theta)$.
- Multiply $\sin(\theta)$ by $\cos(\theta)/\sin(\theta)$ to get $\sin(\theta)\cos(\theta)/\sin(\theta)$.
- Cancel $\sin(\theta)$ in numerator and denominator to obtain $\cos(\theta)$.
- Both sides equal $\cos(\theta)$, so the identity is proven.

Answer: $\cos(\theta) = \cos(\theta)$

2. Prove the identity:

$$\frac{1 - \cos^2(\theta)}{\sin(\theta)} = \sin(\theta)$$

- Use the Pythagorean identity: $1 - \cos^2(\theta) = \sin^2(\theta)$.
- Substitute into the numerator to get $\sin^2(\theta)/\sin(\theta)$.
- Simplify by canceling one factor of $\sin(\theta)$.
- The left side reduces to $\sin(\theta)$, matching the right side.

Answer: $\sin(\theta) = \sin(\theta)$

3. Prove the identity:

$$\sec(\theta) - \sin(\theta)\tan(\theta) = \cos(\theta)$$

- Rewrite $\sec(\theta)$ as $1/\cos(\theta)$ and $\tan(\theta)$ as $\sin(\theta)/\cos(\theta)$.
- The expression becomes $1/\cos(\theta) - \sin^2(\theta)/\cos(\theta)$.
- Combine fractions to get $(1 - \sin^2(\theta))/\cos(\theta)$.
- Apply the Pythagorean identity: $1 - \sin^2(\theta) = \cos^2(\theta)$.
- Simplify $\cos^2(\theta)/\cos(\theta)$ to $\cos(\theta)$.

Answer: $\cos(\theta) = \cos(\theta)$

4. Prove the identity:

$$\frac{1 + \tan^2(\theta)}{\sec^2(\theta)} = 1$$

- Recall the Pythagorean identity: $1 + \tan^2(\theta) = \sec^2(\theta)$.
- Substitute $\sec^2(\theta)$ into the numerator.
- The expression becomes $\sec^2(\theta)/\sec^2(\theta)$.
- This simplifies to 1, matching the right side.

Answer: $1 = 1$



5. Prove the identity:

$$\frac{\cos(\theta)}{1 - \sin(\theta)} = \frac{1 + \sin(\theta)}{\cos(\theta)}$$

→ Multiply the numerator and denominator of the left side by $(1 + \sin(\theta))$.

→ The numerator becomes $\cos(\theta)(1 + \sin(\theta))$.

→ The denominator becomes $1 - \sin^2(\theta)$, which equals $\cos^2(\theta)$.

→ Simplify $\cos(\theta)(1 + \sin(\theta))/\cos^2(\theta)$ by canceling one $\cos(\theta)$.

→ Result is $(1 + \sin(\theta))/\cos(\theta)$, matching the right side.

Answer:
$$\frac{1 + \sin(\theta)}{\cos(\theta)} = \frac{1 + \sin(\theta)}{\cos(\theta)}$$

6. Prove the identity:

$$\tan(\theta) + \cot(\theta) = \sec(\theta)\csc(\theta)$$

→ Rewrite $\tan(\theta)$ as $\sin(\theta)/\cos(\theta)$ and $\cot(\theta)$ as $\cos(\theta)/\sin(\theta)$.

→ Find a common denominator: $\sin(\theta)\cos(\theta)$.

→ Combine: $(\sin^2(\theta) + \cos^2(\theta))/(\sin(\theta)\cos(\theta))$.

→ Apply Pythagorean identity: $\sin^2(\theta) + \cos^2(\theta) = 1$.

→ The expression becomes $1/(\sin(\theta)\cos(\theta)) = \csc(\theta) \cdot \sec(\theta)$.

Answer:
$$\sec(\theta)\csc(\theta) = \sec(\theta)\csc(\theta)$$

7. Prove the identity:

$$\frac{\sin(\theta)}{1 + \cos(\theta)} + \frac{1 + \cos(\theta)}{\sin(\theta)} = 2\csc(\theta)$$

→ Find a common denominator: $\sin(\theta)(1 + \cos(\theta))$.

→ Combine numerators: $\sin^2(\theta) + (1 + \cos(\theta))^2$.

→ Expand $(1 + \cos(\theta))^2 = 1 + 2\cos(\theta) + \cos^2(\theta)$.

→ Add: $\sin^2(\theta) + 1 + 2\cos(\theta) + \cos^2(\theta) = 2 + 2\cos(\theta) = 2(1 + \cos(\theta))$.

→ Divide: $2(1 + \cos(\theta))/[\sin(\theta)(1 + \cos(\theta))] = 2/\sin(\theta) = 2\csc(\theta)$.

Answer:
$$2\csc(\theta) = 2\csc(\theta)$$

8. Prove the identity:

$$(1 - \cos(\theta))(1 + \cos(\theta)) = \sin^2(\theta)$$

→ Expand the product on the left as a difference of squares: $1 - \cos^2(\theta)$.

→ Apply the Pythagorean identity: $1 - \cos^2(\theta) = \sin^2(\theta)$.

→ The left side now equals $\sin^2(\theta)$, matching the right side.

Answer:
$$\sin^2(\theta) = \sin^2(\theta)$$

9. Prove the identity:

$$\frac{\sec^2(\theta) - 1}{\sec^2(\theta)} = \sin^2(\theta)$$

→ Split the fraction: $\sec^2(\theta)/\sec^2(\theta) - 1/\sec^2(\theta)$.

→ Simplify to $1 - \cos^2(\theta)$ since $1/\sec^2(\theta) = \cos^2(\theta)$.

→ Apply Pythagorean identity: $1 - \cos^2(\theta) = \sin^2(\theta)$.

→ The left side equals $\sin^2(\theta)$, matching the right side.

Answer:
$$\sin^2(\theta) = \sin^2(\theta)$$



10. Prove the identity:

$$\csc(\theta) - \sin(\theta) = \cos(\theta)\cot(\theta)$$

→ Rewrite $\csc(\theta)$ as $1/\sin(\theta)$.

→ Combine: $1/\sin(\theta) - \sin(\theta) = (1 - \sin^2(\theta))/\sin(\theta)$.

→ Apply Pythagorean identity: $1 - \sin^2(\theta) = \cos^2(\theta)$.

→ Rewrite as $\cos^2(\theta)/\sin(\theta) = \cos(\theta) \cdot (\cos(\theta)/\sin(\theta))$.

→ This equals $\cos(\theta) \cdot \cot(\theta)$, matching the right side.

Answer: $\cos(\theta)\cot(\theta) = \cos(\theta)\cot(\theta)$

